Resonant cascaded Kerr nonlinearity in $\chi^{(2)}$ metasurfaces: supplemental document

Derivation of the cascaded susceptibility of a resonator, its uniform-medium limit, its frequency response and strength–bandwidth relation, the thresholds, the host-material parameters, the coupled-mode simulations, the singly resonant metasurface and the microring model.

1. COUPLED-MODE EQUATIONS AND THE CASCADED SUSCEPTIBILITY

A. Conventions and modes

Real fields are written as $\mathcal{E} = \mathbf{E}e^{-i\omega t} + \text{c.c.}$ without a factor $\frac{1}{2}$, so that the time average of $\mathcal{E}^2$ is $2|\mathbf{E}|^2$. With $\mathbf{E}_1$ and $\mathbf{E}_2$ the complex amplitudes at ω_1 and $2\omega_1$, the second-harmonic, difference-frequency and self-phase-modulation polarisations are [1]

$$P_i(2\omega_1) = \varepsilon_0\chi^{(2)}_{ijk}E_{1,j}E_{1,k}, \quad P_i(\omega_1) = 2\varepsilon_0\chi^{(2)}_{ijk}E_{2,j}E^*_{1,k}, \quad P_i(\omega_1) = 3\varepsilon_0\chi^{(3)}_{ijkl}E_{1,j}E_{1,k}E^*_{1,l}, \tag{S1}$$

where the factors 2 and 3 count the distinct orderings of the input frequencies. Kleinman symmetry makes $\chi^{(2)}_{ijk}$ invariant under permutation of its indices, so one real tensor $\chi^{(2)}_{ijk}(\mathbf{r})$ describes both second-order processes.

Mode n has the resonance frequency $\bar{\omega}_n$, with $\bar{\omega}_1 = \omega_a$ for the pump mode and $\bar{\omega}_2 = \omega_2$ for the second-harmonic mode, and satisfies $\nabla\times\nabla\times\mathbf{f}_n = (\bar{\omega}_n^2/c^2)\varepsilon_r(\mathbf{r},\bar{\omega}_n)\mathbf{f}_n$ with $\mathbf{h}_n = -i\nabla\times\mathbf{f}_n/(\mu_0\bar{\omega}_n)$. We define

$$N_n = \frac{1}{2}\int\left(\varepsilon_0\tilde{\varepsilon}_n|\mathbf{f}_n|^2 + \mu_0|\mathbf{h}_n|^2\right)d^3r, \qquad \tilde{\varepsilon}_n = \left.\frac{\partial(\omega\varepsilon_r)}{\partial\omega}\right|_{\bar{\omega}_n}, \tag{S2}$$

so that, by Brillouin's expression for the energy in a dispersive medium, the field $a_n\mathbf{f}_n e^{-i\bar{\omega}_n t} + \text{c.c.}$ stores the energy $W_n = 2N_n|a_n|^2$. For an eigenmode $\int\varepsilon_0\varepsilon_r|\mathbf{f}_n|^2d^3r = \int\mu_0|\mathbf{h}_n|^2d^3r$, and $N_n = \int\varepsilon_0\varepsilon_r|\mathbf{f}_n|^2d^3r$ in a nondispersive medium. The susceptibilities and overlaps below do not depend on the normalisation of the modes.

B. Projection onto a mode

Write the field near frequency ω as $\mathcal{E} = a(t)\mathbf{f}_n(\mathbf{r})e^{-i\omega t} + \text{c.c.}$ with a slowly varying amplitude a, and the nonlinear polarisation in the same band as $\mathbf{P}e^{-i\omega t}$ plus its complex conjugate. The positive-frequency part of $\nabla\times\nabla\times\mathcal{E} + c^{-2}\partial_t^2\mathcal{D}_{\text{lin}}/\varepsilon_0 = -\mu_0\partial_t^2\mathcal{P}$, where $\mathcal{D}_{\text{lin}}$ is the linear displacement field and $\mathcal{P}$ the nonlinear polarisation, contains the operator $(\omega+i\partial_t)^2\varepsilon_r(\omega+i\partial_t)$. To first order in ∂_t and in $\omega-\bar{\omega}_n$ it equals $\bar{\omega}_n^2\varepsilon_r(\bar{\omega}_n) - \bar{\omega}_n(\varepsilon_r+\tilde{\varepsilon}_n)(\bar{\omega}_n-\omega) + i\bar{\omega}_n(\varepsilon_r+\tilde{\varepsilon}_n)\partial_t$. Neglecting $\dot{\mathbf{P}}$ against $\omega\mathbf{P}$ and projecting onto $\mathbf{f}_n^*$ gives

$$\dot{a} = i(\omega-\bar{\omega}_n)\,a + \frac{i\omega}{2N_n}\int\mathbf{f}_n^*\cdot\mathbf{P}\,d^3r, \tag{S3}$$

since $\int\varepsilon_0(\varepsilon_r+\tilde{\varepsilon}_n)|\mathbf{f}_n|^2d^3r = 2N_n$. For a static perturbation in a nondispersive medium this is the first-order frequency shift $-(\bar{\omega}_n/2)\int\mathbf{E}^*\cdot\delta\mathbf{P}\,d^3r/\int\varepsilon_0\varepsilon_r|\mathbf{E}|^2d^3r$ [2]. Radiation and absorption are included by giving $\bar{\omega}_n$ the imaginary part $-\gamma_n/2$, and the pump detuning is $\delta_1 = \omega_1 - \omega_a$. The pump mode is driven by the term $(\gamma_1^{\text{e}}/2N_1)^{1/2}s$, where $|s|^2$ is the incident power and γ_1^{e} is the decay rate into the channel of the incident wave, half the radiative rate for a structure that radiates equally to both sides.

C. Coupling constants

With $\mathbf{E}_1 = a_1\mathbf{f}_1$ and $\mathbf{E}_2 = a_2\mathbf{f}_2$, the three sources of Eq. Eq. (S1) give, through Eq. Eq. (S3),

$$\int \mathbf{f}_2^* \cdot \mathbf{P}(2\omega_1)\, d^3r = \varepsilon_0 M a_1^2, \qquad \beta = \frac{\varepsilon_0\omega_1 M}{N_2}, \tag{S4}$$

$$\int \mathbf{f}_1^* \cdot \mathbf{P}_{\mathrm{DFG}}(\omega_1)\, d^3r = 2\varepsilon_0 M^* a_1^* a_2, \qquad \beta_1 = \frac{\varepsilon_0\omega_1 M}{N_1}, \tag{S5}$$

$$\int \mathbf{f}_1^* \cdot \mathbf{P}_{\mathrm{SPM}}(\omega_1)\, d^3r = 3\varepsilon_0 K|a_1|^2 a_1, \qquad \alpha_3 = \frac{3\varepsilon_0\omega_1}{2N_1}K, \tag{S6}$$

with $M = \int \chi^{(2)}_{ijk} f^*_{2,i} f_{1,j} f_{1,k}\, d^3r$ and $K = \int \chi^{(3)}_{ijkl} f^*_{1,i} f_{1,j} f_{1,k} f^*_{1,l}\, d^3r$. The second line uses Kleinman symmetry, $\int \chi^{(2)}_{ijk} f^*_{1,i} f_{2,j} f^*_{1,k}\, d^3r = M^*$. These are the coefficients of Eqs. (1) and (2) of the main text. In the absence of loss and drive,

$$\frac{d}{dt}\left(N_1|a_1|^2 + N_2|a_2|^2\right) = -2\,\mathrm{Im}\left(\varepsilon_0\omega_1 M^* a_1^{*2} a_2\right) + 2\,\mathrm{Im}\left(\varepsilon_0\omega_1 M^* a_1^{*2} a_2\right) = 0, \tag{S7}$$

so the stored energy $W_1 + W_2$ is conserved, which requires the factor 2 of the difference-frequency polarisation.

D. Steady-state elimination

If a_1^2 varies slowly on the time scale $|\Delta\omega + i\gamma_2/2|^{-1}$, Eq. (2) of the main text gives $a_2 = -\beta a_1^2/(\Delta\omega + i\gamma_2/2)$ with $\Delta\omega = 2\omega_1 - \omega_2$. The term $i\beta_1^* a_1^* a_2$ of Eq. (1) of the main text becomes $i\alpha_{\mathrm{casc}}|a_1|^2 a_1$ with

$$\alpha_{\mathrm{casc}} = -\frac{\beta_1^*\beta}{\Delta\omega + i\gamma_2/2} = -\frac{\varepsilon_0^2\omega_1^2|M|^2}{N_1 N_2(\Delta\omega + i\gamma_2/2)}. \tag{S8}$$

Integration of the two-mode equations reproduces Eq. Eq. (S8) [Fig. S1(a)]. The real part of α_{casc} shifts the pump resonance, and its imaginary part,

$$\mathrm{Im}\,\alpha_{\mathrm{casc}} = \frac{\varepsilon_0^2\omega_1^2|M|^2\,\gamma_2/2}{N_1 N_2[\Delta\omega^2 + (\gamma_2/2)^2]}, \tag{S9}$$

reduces the pump energy at the rate $2\,\mathrm{Im}\,\alpha_{\mathrm{casc}}|a_1|^2 W_1$, which in steady state equals the power $\gamma_2 W_2$ lost by the second-harmonic mode.

The native susceptibility $\chi^{(3)}_{\mathrm{nat}}$ of the second-order material is defined by $K = \chi^{(3)}_{\mathrm{nat}}\int_{\mathrm{NL}}|\mathbf{f}_1|^4 d^3r$, so that

$$\alpha_3 = \frac{3\varepsilon_0\omega_1}{2N_1}\,\chi^{(3)}_{\mathrm{nat}}\int_{\mathrm{NL}}|\mathbf{f}_1|^4 d^3r, \tag{S10}$$

where $\int_{\mathrm{NL}}$ extends over that material. For an isotropic response and a mode that is linearly polarised everywhere in the material, $\chi^{(3)}_{\mathrm{nat}} = \chi^{(3)}_{xxxx}$. Other materials in the cell are assumed to have a negligible $\chi^{(3)}$, and the native response is taken to be lossless. Defining $\chi^{(3)}_{\mathrm{casc}}$ in the same way through α_{casc} gives

$$\chi^{(3)}_{\mathrm{casc}} = -\frac{2\varepsilon_0\omega_1}{3N_2}\,\frac{|M|^2}{(\Delta\omega + i\gamma_2/2)\int_{\mathrm{NL}}|\mathbf{f}_1|^4 d^3r}, \tag{S11}$$

in which N_1 cancels and N_2 cancels the normalisation of $\mathbf{f}_2$ in $|M|^2$. With the overlap $\eta_{\mathrm{geom}} = \varepsilon_0 n_2^2|M|^2/(|\chi^{(2)}|^2 N_2\int_{\mathrm{NL}}|\mathbf{f}_1|^4 d^3r)$, Eq. Eq. (S11) takes the form of Eq. (3) of the main text, and $\alpha_{\mathrm{casc}}/\alpha_3 = \chi^{(3)}_{\mathrm{casc}}/\chi^{(3)}_{\mathrm{nat}}$.

E. Bound on the overlap

Let $|\chi^{(2)}|$ be the largest value of $|\chi^{(2)}_{ijk} u_i^* v_j v_k|$ over complex unit vectors $\mathbf{u}, \mathbf{v}$ and positions in the material. It equals $2d_{33}$ for $LiNbO_3$ and AlN and $(2/\sqrt{3})\,2d_{14}$ for zincblende crystals, where the maximum is reached for fields along [111]. Let n_2^2 be the smallest principal value of ε_r of the material at $2\omega_1$, which is n_e^2 for $LiNbO_3$. The Cauchy–Schwarz inequality gives

$$|M|^2 \le |\chi^{(2)}|^2 \int_{\mathrm{NL}}|\mathbf{f}_2|^2 d^3r \int_{\mathrm{NL}}|\mathbf{f}_1|^4 d^3r. \tag{S12}$$

For normal dispersion $\tilde{\varepsilon}_2 \geq \varepsilon_r$, so that $N_2 \geq \int \varepsilon_0 \varepsilon_r |\mathbf{f}_2|^2 d^3r \geq \varepsilon_0 n_2^2 \int_{\mathrm{NL}} |\mathbf{f}_2|^2 d^3r$, and therefore $\eta_{\mathrm{geom}} \leq 1$. Equality requires that $\mathbf{f}_2$ vanish outside the material, that $\chi^{(2)}_{ijk} f_{1,j} f_{1,k}$ be parallel to $\mathbf{f}_2$ with a constant ratio and magnitude $|\chi^{(2)}||\mathbf{f}_1|^2$, and that $\tilde{\varepsilon}_2 = \varepsilon_r = n_2^2$ wherever $\mathbf{f}_2$ is nonzero. The uniform medium of Section 3 meets these conditions when dispersion within the second-harmonic band is neglected. When both fields are uniform in the material and aligned with the maximising direction, η_{geom} is the fraction of the electric energy of the second-harmonic mode that lies in the material.

2. GREEN'S-FUNCTION FORM OF THE CASCADE

At leading order in the nonlinearity the second-harmonic field obeys

$$\left[\nabla \times \nabla \times - k_0^2 \varepsilon_r(\mathbf{r}, 2\omega_1)\right] \mathbf{E}_2 = k_0^2 \chi^{(2)} : \mathbf{E}_1 \mathbf{E}_1, \tag{S13}$$

with $k_0 = 2\omega_1/c$ the vacuum wavenumber at $2\omega_1$, so that

$$E_{2,j}(\mathbf{r}) = k_0^2 \int G_{jl}(\mathbf{r}, \mathbf{r}') \chi^{(2)}_{lmn}(\mathbf{r}') E_{1,m}(\mathbf{r}') E_{1,n}(\mathbf{r}')\, d^3r', \tag{S14}$$

with G the Green's function of the operator on the left. Inserting this field into the difference-frequency polarisation of Eq. Eq. (S1) gives

$$P_i^{\mathrm{casc}}(\mathbf{r}) = 2\varepsilon_0 k_0^2 E^*_{1,k}(\mathbf{r}) \int \chi^{(2)}_{ijk}(\mathbf{r})\, G_{jl}(\mathbf{r}, \mathbf{r}')\, \chi^{(2)}_{lmn}(\mathbf{r}')\, E_{1,m}(\mathbf{r}') E_{1,n}(\mathbf{r}')\, d^3r'. \tag{S15}$$

The polarisation at $\mathbf{r}$ depends on the pump field at every point that the second-harmonic field connects to $\mathbf{r}$, so the cascade is nonlocal. In a homogeneous crystal of length L with $|\Delta k| L \gg 1$ the kernel reduces to the local bulk response of Section 3.

Near an isolated mode the Green's function is dominated by its pole. Expanding the operator of Eq. Eq. (S13) about ω_2 as in Section 1.B gives

$$G_{jl}(\mathbf{r}, \mathbf{r}'; 2\omega_1) \simeq - \frac{\varepsilon_0 c^2}{4\omega_1 N_2} \frac{f_{2,j}(\mathbf{r})\, f^*_{2,l}(\mathbf{r}')}{\Delta\omega + i\gamma_2/2} + G^{\mathrm{bg}}_{jl}, \tag{S16}$$

with a nonresonant background G^{bg}. For a leaky mode the product $f_{2,j} f^*_{2,l}/N_2$ is replaced by the quasinormal-mode residue, which contains the unconjugated product of the mode with its reciprocal partner and a complex normalisation [3, 4]. For a standing-wave mode with $Q_2 = \omega_2/\gamma_2 \gg 1$ the field can be taken real inside the resonator and the residue reduces to Eq. Eq. (S16); the imaginary part of the normalisation adds a correction of order $1/Q_2$ that makes the response asymmetric about $\Delta\omega = 0$. With $\mathbf{E}_1 = a_1 \mathbf{f}_1$, Eq. Eq. (S16) gives $\mathbf{E}_2 = a_2 \mathbf{f}_2$ with $a_2 = -\beta a_1^2/(\Delta\omega + i\gamma_2/2)$, the steady state of Eq. (2) of the main text, and Eq. Eq. (S15) becomes

$$P_i^{\mathrm{casc}} = - \frac{2\varepsilon_0^2 \omega_1 M |a_1|^2 a_1}{N_2(\Delta\omega + i\gamma_2/2)} \chi^{(2)}_{ijk} f_{2,j} f^*_{1,k}. \tag{S17}$$

Its projection onto $\mathbf{f}_1^*$ is $-2\varepsilon_0^2 \omega_1 |M|^2 |a_1|^2 a_1 / [N_2(\Delta\omega + i\gamma_2/2)]$. Equating it to the projection $3\varepsilon_0 \chi^{(3)}_{\mathrm{casc}} |a_1|^2 a_1 \int_{\mathrm{NL}} |\mathbf{f}_1|^4 d^3r$ of a native polarisation returns Eq. Eq. (S11).

3. UNIFORM-MEDIUM LIMIT AND A LAYERED-CELL TEST

In a uniform isotropic medium with refractive indices n_1 at ω_1 and n_2 at $2\omega_1$, a pump plane wave $A\hat{\mathbf{x}} e^{ik_1 z}$ with $k_1 = n_1\omega_1/c$ drives the source $\varepsilon_0 A^2 \mathbf{p} e^{2ik_1 z}$, where $p_m = \chi^{(2)}_{mxx}$. For the transverse part of $\mathbf{p}$, Eq. Eq. (S13) gives the bound wave

$$\mathbf{E}_{2\perp} = \frac{\mathbf{p}_\perp A^2}{n_1^2 - n_2^2}. \tag{S18}$$

For $\mathbf{p}_\perp = |\chi^{(2)}| \hat{\mathbf{x}}$ the difference-frequency polarisation along $\hat{\mathbf{x}}$ is $2\varepsilon_0 |\chi^{(2)}| E_{2,x} A^*$, and

$$\chi^{(3)}_{\mathrm{casc,bulk}} = - \frac{2|\chi^{(2)}|^2}{3(n_2^2 - n_1^2)} \simeq - \frac{2\omega_1 |\chi^{(2)}|^2}{3 c n_2 \Delta k}, \qquad \Delta k = \frac{2\omega_1 (n_2 - n_1)}{c}, \tag{S19}$$

where the second form holds to first order in $n_2 - n_1$. The longitudinal component p_z does not propagate, and $\nabla \cdot \mathbf{D} = 0$ gives $E_{2,z} = -p_z A^2/n_2^2$, which adds $-2p_z^2/(3n_2^2)$ to the cascaded susceptibility. This term does not depend on $\Delta\omega$, is smaller than the resonant one by a factor of order $|\Delta\omega|/\omega_1$, and vanishes for a transverse $\mathbf{p}$.

In a periodic box of volume V filled with the medium, and neglecting dispersion within the second-harmonic band, $\mathbf{f}_1 = \hat{\mathbf{x}} e^{ik_1 z}/(\varepsilon_0 n_1^2 V)^{1/2}$ couples only to $\mathbf{f}_2 = \hat{\mathbf{x}} e^{2ik_1 z}/(\varepsilon_0 n_2^2 V)^{1/2}$, whose frequency is $\omega_2 = 2\omega_1 n_1/n_2$, and $N_1 = N_2 = 1$. Then $|M|^2 = |\chi^{(2)}|^2/(\varepsilon_0^3 n_1^4 n_2^2 V)$, $\int |\mathbf{f}_1|^4 d^3r = 1/(\varepsilon_0^2 n_1^4 V)$, $\eta_{\mathrm{geom}} = 1$, and Eq. Eq. (S11) with $\Delta\omega = 2\omega_1(n_2 - n_1)/n_2$ and $\gamma_2 = 0$ gives $-|\chi^{(2)}|^2/[3n_2(n_2 - n_1)]$, which agrees with Eq. Eq. (S19) to first order in $n_2 - n_1$. The residual comes from linearising $\omega_2^2 - 4\omega_1^2$ about the pole; with the full denominator of the mode expansion the two expressions coincide. Dispersion within the second-harmonic band changes N_2 to n_{g2}/n_2 and $\Delta\omega$ to $c\Delta k/n_{g2}$, with n_{g2} the group index at $2\omega_1$, so that $\eta_{\mathrm{geom}} = n_2/n_{g2}$ and $\chi^{(3)}_{\mathrm{casc}}$ is unchanged.

To test Eq. Eq. (S11) in an inhomogeneous structure, we take a one-dimensional Bloch-periodic cell of period Λ in which a layer of refractive index 2.2 and thickness 0.45Λ carries $\chi^{(2)}$ and the rest has refractive index 1.45. The pump mode at Bloch wavevector $0.37\pi/\Lambda$ is computed by finite differences, the second-harmonic field that it drives is found from the full Helmholtz equation without a modal expansion, and the resulting difference-frequency polarisation is projected onto the pump mode. The mismatch $\Delta\omega$ is varied by scaling the permittivities at $2\omega_1$ by $1 + d$ in the nonlinear layer and by $1 + 0.3d$ in the other layer. The ratio of this result to Eq. Eq. (S11) approaches 1 linearly in $\Delta\omega/2\omega_1$, mainly because of the linearised pole, as in the uniform medium; the nonresonant second-harmonic modes reduce the deviation. It differs from 1 by 4×10^{-4} at $\Delta\omega/2\omega_1 = 10^{-3}$ [Fig. S1(b)].

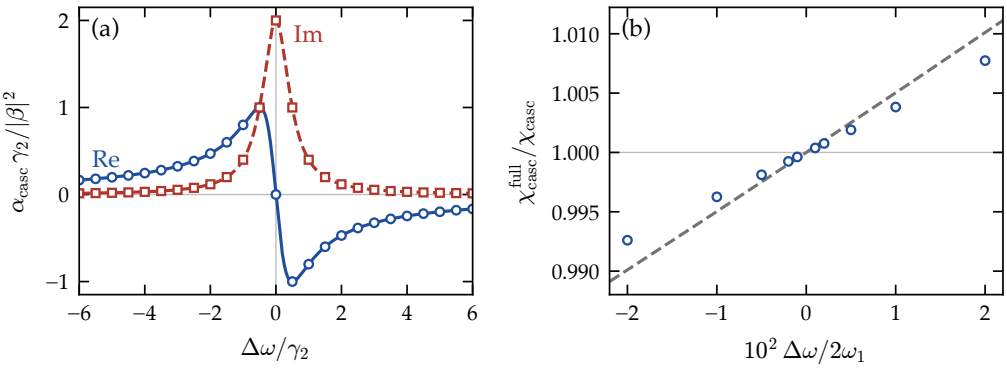


Fig. S1. (a) Real (solid line, circles) and imaginary (dashed line, squares) parts of the cascaded coefficient α_{casc} of the pump mode, in units of $|\beta|^2/\gamma_2$ with $N_1 = N_2$, from steady-state integration of the two-mode equations (markers) and from Eq. Eq. (S8) (lines) versus $\Delta\omega/\gamma_2$, for $\gamma_1 = 10\gamma_2$, $\gamma_1^{\mathrm{e}} = \gamma_1/2$, $\delta_1 = 0$, $\alpha_3 = 0$ and a drive $|s|^2 = 10^{-4}\gamma_2^3/|\beta|^2$. (b) Cascaded susceptibility of the two-layer Bloch cell of Section 3 (indices 2.2 and 1.45, nonlinear layer 0.45Λ, Bloch wavevector $0.37\pi/\Lambda$, 1200 grid points per period) from the full driven second-harmonic field (circles), divided by Eq. Eq. (S11), versus $\Delta\omega/2\omega_1$. The dashed line is the single-mode pole without linearisation, $1/(1 - \Delta\omega/4\omega_1)$.

4. RESPONSE TIME, LOSS AND THRESHOLDS

A. Response to a modulated pump

Integrating Eq. (2) of the main text without the steady-state assumption gives

$$a_2(t) = i\beta \int_{-\infty}^{t} e^{(i\Delta\omega - \gamma_2/2)(t-t')} a_1^2(t')\, dt'. \tag{S20}$$

For $a_1^2(t) = \int \tilde{A}(\Omega) e^{-i\Omega t} d\Omega$, so that Ω is measured from $2\omega_1$, this is $a_2 = -\beta \int \tilde{A}(\Omega) e^{-i\Omega t}/(\Delta\omega + \Omega + i\gamma_2/2)\, d\Omega$. Each component $\tilde{A}(\Omega)$ of a_1^2 therefore enters Eq. (1) of the main text with the

cascaded susceptibility

$$\chi^{(3)}_{\mathrm{casc}}(\Omega) = -\frac{C}{\Delta\omega + \Omega + i\gamma_2/2}, \qquad C = \frac{2\omega_1 |\chi^{(2)}|^2 \eta_{\mathrm{geom}}}{3 n_2^2}. \tag{S21}$$

A real modulation of the pump intensity at frequency Ω contains both components $\pm\Omega$ and samples $\chi^{(3)}_{\mathrm{casc}}(\pm\Omega)$. The response has a single pole at $\Omega = -\Delta\omega - i\gamma_2/2$, in the lower half-plane, and for any Q_2 and $\Delta\omega$ it obeys

$$|\chi^{(3)}_{\mathrm{casc}}(0)|\,|\Delta\omega + i\gamma_2/2| = C, \qquad \int_{-\infty}^{\infty} \mathrm{Im}\,\chi^{(3)}_{\mathrm{casc}}(\Omega)\, d\Omega = \pi C. \tag{S22}$$

The first relation fixes the product of the static susceptibility and the distance of the pole from the origin, which limits the quasi-static bandwidth. The second, a sum rule over the modulation frequency, states that the integrated nonlinear absorption does not depend on the linewidth. With $\Delta f \le |\Delta\omega + i\gamma_2/2|/2\pi$ the first relation gives Eq. (5) of the main text. At $\Delta\omega = 2\gamma_2$, $|\chi^{(3)}_{\mathrm{casc}}(\Omega)/\chi^{(3)}_{\mathrm{casc}}(0)|$ lies between 0.81 and 1.30 for $|\Omega| \le \gamma_2/2$.

B. Nonlinear loss and stability

From Eq. Eq. (S11), $\mathrm{Im}\,\chi^{(3)}_{\mathrm{casc}}/\mathrm{Re}\,\chi^{(3)}_{\mathrm{casc}} = -\gamma_2/(2\Delta\omega)$, so the loss-to-Kerr ratio of the cascade is $\rho = \gamma_2/(2|\Delta\omega|)$. The second-harmonic energy is $W_2 = 2N_2|a_2|^2$, and

$$\frac{W_2}{W_1} = \frac{|\alpha_{\mathrm{casc}}|\,|a_1|^2}{|\Delta\omega + i\gamma_2/2|}. \tag{S23}$$

This ratio equals $S/|\Delta\omega|$, where $S = |\mathrm{Re}\,\alpha_{\mathrm{casc}}||a_1|^2$ is the cascaded shift of the pump resonance. In a continuous-wave steady state the elimination of a_2 is exact at any power, and $\mathrm{Im}\,\alpha_{\mathrm{casc}}$ accounts for the power transferred to the second harmonic. The steady state can, however, lose stability through a Hopf bifurcation to self-pulsing, as in doubly resonant second-harmonic generation [5]. In the two-mode equations this happens once S exceeds $|\Delta\omega|$ by a factor that depends on γ_1 and δ_1. The factor is about 1.1 for the microring of Section 8 and 2.9 for $\gamma_1 = 10\gamma_2$, both at $\delta_1 = 0$, and reaches about 10 at large pump detuning.

Equations (1) and (2) of the main text keep only the native self-phase modulation of the pump, which shifts its resonance by $\alpha_3|a_1|^2 = S/R$, with $R = |\mathrm{Re}\,\alpha_{\mathrm{casc}}|/\alpha_3$ as in Eq. (6) of the main text. For a field that is uniform in the second-order material, the native cross-phase modulation, with the factor 6 of Ref. [1], shifts the second-harmonic mode by about $4S/R$ and the pump mode by about $2S^2/(R|\Delta\omega|)$, and the native self-phase modulation of the second harmonic shifts its mode by about $2S^2/(R|\Delta\omega|)$. The two shifts of the second-harmonic mode change $\Delta\omega$ and must be small compared with $|\Delta\omega|$, which requires $S \ll |\Delta\omega|R/4$ and $S \ll |\Delta\omega|(R/2)^{1/2}$; for $R \gtrsim 8$ the second condition is the stricter. For $S < 2|\Delta\omega|$ the cross-phase term is the larger of the two; at the folds of Fig. 3(c) of the main text, where S is $0.46|\Delta\omega|$ and $0.78|\Delta\omega|$ and $R = 20$, it amounts to at most $0.16|\Delta\omega|$.

C. Bistability

Write the total Kerr coefficient $\alpha_3 + \alpha_{\mathrm{casc}}$ as $\alpha = \alpha' + i\alpha''$ with $\alpha'' \ge 0$, and let $\rho_{\mathrm{tot}} = \alpha''/|\alpha'|$ and $y = |a_1|^2$. Then $\rho_{\mathrm{tot}} = \rho R/|R \mp 1|$, with the upper sign for $\Delta\omega > 0$, where the cascaded and native terms have opposite signs; for $R \gg 1$, $\rho_{\mathrm{tot}} \simeq \rho$. The steady state of Eq. (1) of the main text gives

$$\frac{\gamma_1^{\mathrm{e}}}{2N_1}|s|^2 = y\left[(\delta_1 + \alpha' y)^2 + (\gamma_1/2 + \alpha'' y)^2\right], \tag{S24}$$

and the response is bistable when $d|s|^2/dy < 0$ for some $y > 0$. The derivative is the quadratic $3|\alpha|^2 y^2 + 4(\alpha'\delta_1 + \alpha''\gamma_1/2)y + \delta_1^2 + \gamma_1^2/4$, which has positive roots only if the vectors (α', α'') and $(\delta_1, \gamma_1/2)$ make an angle larger than 150°. The first vector makes an angle $\arctan\rho_{\mathrm{tot}}$ with the α' axis and the second lies in the upper half-plane, so bistability is possible only for $\arctan\rho_{\mathrm{tot}} < 30°$, that is $\rho_{\mathrm{tot}} < 1/\sqrt{3}$. It then requires a pump detuning of the opposite sign to α' with

$$\frac{2|\delta_1|}{\gamma_1} > \cot\left(30° - \arctan\rho_{\mathrm{tot}}\right), \tag{S25}$$

which is $\sqrt{3}$ for $\rho_{\rm tot}=0$ and 3.50 for $\rho_{\rm tot}=1/4$. Since $\Delta\omega$ is defined with the pump frequency, the resonator must be designed with the cold mismatch $2\omega_a-\omega_2=\Delta\omega-2\delta_1$.

We take as the bistability threshold the input power at which the lower branch switches up, the larger of the two fold powers, minimised over δ_1. Because the cascaded loss raises the required detuning, the threshold falls by less than the Kerr coefficient. With $\alpha_3=|\mathrm{Re}\,\alpha_{\rm casc}|/R$ and $\Delta\omega=-2\gamma_2$, so that the cascaded and native terms add and $\rho=1/4$, the threshold with the cascade is lower than that of the native response alone by a factor $(1+R)/4.7$ for $R=10$, $(1+R)/5.2$ for $R=20$ and $(1+R)/5.8$ for $R\to\infty$. The last value is the ratio of the thresholds of a lossless Kerr term and of one with $\rho_{\rm tot}=1/4$.

D. Parametric gain

In a resonator with several pump-band modes, a pair of sidebands $a_\pm$ at frequencies $\omega_1\pm\nu$, in modes detuned from them by $\delta_\pm$, obeys in the effective Kerr description

$$\dot a_\pm=\big(i\delta_\pm-\tfrac{\gamma_1}{2}+2i\alpha y\big)a_\pm+i\alpha a_1^2a_\mp^*. \tag{S26}$$

For symmetric detunings the largest growth rate, reached when $\delta_\pm+2\alpha' y=0$, is $(|\alpha|-2\alpha'')y-\gamma_1/2$. Gain is possible only if $|\alpha|>2\alpha''$, which is again $\rho_{\rm tot}<1/\sqrt{3}$, and the threshold is $y=\gamma_1/[2|\alpha'|(\sqrt{1+\rho_{\rm tot}^2}-2\rho_{\rm tot})]$. With a single second-harmonic mode the cross terms of the sidebands carry the cascaded coefficient at $\Delta\omega\pm\nu$, so this description requires $|\nu|\ll|\Delta\omega+i\gamma_2/2|$. In a multimode resonator the cross terms couple the sidebands through other second-harmonic modes, and the threshold must be computed with those modes included (Section 8).

5. HOST MATERIALS

At $\Delta\omega=2\gamma_2$ and with $\gamma_2=\omega_2/Q_2\simeq2\omega_1/Q_2$, Eq. Eq. (S11) gives

$$R=\frac{|\mathrm{Re}\,\chi^{(3)}_{\rm casc}|}{\chi^{(3)}_{\rm nat}}=\frac{Q_2\eta_{\rm geom}F}{6}\cdot\frac{1}{1+(\gamma_2/2\Delta\omega)^2}=\frac{Q_2\eta_{\rm geom}F}{6.375},\qquad F=\frac{|\chi^{(2)}|^2}{n_2^2\chi^{(3)}_{\rm nat}}. \tag{S27}$$

The main text quotes the approximation $Q_2\eta_{\rm geom}F/6$, and $R=1$ at $Q_2=6/(\eta_{\rm geom}F)$.

Table S1 lists the constants for a pump at 1.55 µm. The second-order coefficients are scaled to this wavelength with Miller's rule [6]. For congruent $LiNbO_3$ we start from $d_{33}=19.5$ pm/V at 1.313 µm [7], and for bulk AlN from $d_{33}=4.3$ pm/V at 1.03 µm [8]. For GaP the starting value is $d_{14}=70.6$ pm/V at 1.064 µm [7]; its second harmonic lies above the indirect gap, so the scaled value is an upper estimate. For $Al_{0.2}Ga_{0.8}As$ we use the ratio $d_{14}/d_{33}=5.01$ of GaAs to $LiNbO_3$ at 4.1 µm [9] and scale it in wavelength and composition, which gives $d_{14}=92.6$ pm/V. The refractive indices are from Refs. [10–13]; the pump and the second harmonic are extraordinary in $LiNbO_3$ and AlN, and n_2, the smallest principal index at $2\omega_1$, is n_e for $LiNbO_3$ and n_o for AlN. The native Kerr indices are $n_2^I=1.8\times10^{-19}$ m^2/W for $LiNbO_3$, the value used for microresonators [14], 1.1×10^{-17} m^2/W for GaP [15], 3.5×10^{-19} m^2/W for AlN [16] and 2.6×10^{-17} m^2/W for AlGaAs [17]. They are converted with $\chi^{(3)}_{\rm nat}=4\varepsilon_0cn_1^2n_2^I/3$, where n_1 is the index at the pump.

The native index of $LiNbO_3$ is the least certain input. The value above was inferred from the parametric-oscillation threshold of a microring with ordinary polarisation [18]. Cross-phase modulation in a thin-film ring at 1550 nm gives 1.74×10^{-19} m^2/W for extraordinary and 1.61×10^{-19} m^2/W for ordinary polarisation, with a cascaded contribution of opposite sign estimated below 10% and not subtracted [19]. A Z-scan at 1.064 µm gives 0.91×10^{-19} m^2/W [20], a value that contains the negative cascaded contribution of the bulk crystal; without that contribution the index is 3.9×10^{-19} m^2/W [21, 22]. Equation Eq. (S19) accounts for the difference: with $d_{33}=25.2$ pm/V at 1.064 µm [7] and the extraordinary indices of Ref. [10], the bulk cascade contributes -3.0×10^{-19} m^2/W, and the sum with 3.9×10^{-19} m^2/W is 0.89×10^{-19} m^2/W. With the two-band dispersion of Ref. [23] we estimate about 3.2×10^{-19} m^2/W at 1.55 µm. For n_2^I between 1.8 and 3.2×10^{-19} m^2/W and d_{33} between 18.8 and 22.7 pm/V (Miller's rule from the two wavelengths of Ref. [7]), R of $LiNbO_3$ at $Q_2=4000$ and $\eta_{\rm geom}=0.3$ lies between 11 and 30. For GaP, AlN and AlGaAs the reported spreads give F between 2×10^{-3} and 9×10^{-3}, so the three cannot be ranked against one another. Their absorption differs. AlN absorbs at neither wavelength, and GaP has two-photon absorption at the second harmonic [24]. The second harmonic of $Al_{0.2}Ga_{0.8}As$ lies 74 meV below the band gap of 1.67 eV [25], where the absorption

tail adds loss, and the two-photon absorption of one pump and one second-harmonic photon, with $0.8 + 1.6$ eV above the gap, adds a pump loss that grows with the second-harmonic intensity. GaAs is not listed because its second harmonic at 775 nm lies above the band gap.

Table S1. Host Constants for a Pump at 1.55 µm and the Ratio $R = Q_2\eta_{\mathrm{geom}}F/6$ at $Q_2 = 4000$, $\Delta\omega = 2\gamma_2$ and $\eta_{\mathrm{geom}} = 0.3$

Host	$\lvert\chi^{(2)}\rvert$ (pm/V)	n_1	n_2	n_2^I (m^2/W)	$\chi^{(3)}_{\mathrm{nat}}$ (m^2/V^2)	F	R	Q_2 at $R = 1$
$LiNbO_3$	37.5	2.14	2.18	1.8×10^{-19}	2.9×10^{-21}	0.10	20	200
GaP	130	3.05	3.20	1.1×10^{-17}	3.6×10^{-19}	0.0045	0.91	4.4×10^3
AlN	8.1	2.16	2.14	3.5×10^{-19}	5.8×10^{-21}	0.0025	0.49	8.1×10^3
$Al_{0.2}Ga_{0.8}As$	214	3.27	3.53	2.6×10^{-17}	9.8×10^{-19}	0.0037	0.75	5.4×10^3

6. COUPLED-MODE SIMULATIONS

Equations, integrator, tolerances, run lengths and convergence criterion for Figs. 3 and 4 of the main text and Fig. S1, and the time-domain model for pulsed excitation used in Fig. 4(d) of the main text.

7. SINGLY RESONANT METASURFACE

Without a pump resonance the pump field in the structure is $E_{\mathrm{in}}\mathbf{e}_1(\mathbf{r})$, with E_{in} the incident amplitude and $\mathbf{e}_1$ the linear field per unit incident amplitude. Equation (S3) for the second-harmonic mode gives, in steady state,

$$a_2 = -\frac{\varepsilon_0\omega_1\widetilde{M}E_{\mathrm{in}}^2}{N_2(\Delta\omega + i\gamma_2/2)}, \qquad \widetilde{M} = \int \chi^{(2)}_{ijk} f^*_{2,i} e_{1,j} e_{1,k}\, d^3r. \tag{S28}$$

The projections of the cascaded and native polarisations onto $\mathbf{e}_1^*$ are

$$-\frac{2\varepsilon_0^2\omega_1|\widetilde{M}|^2}{N_2(\Delta\omega + i\gamma_2/2)}\,|E_{\mathrm{in}}|^2E_{\mathrm{in}} \quad \text{and} \quad 3\varepsilon_0\chi^{(3)}_{\mathrm{nat}}|E_{\mathrm{in}}|^2E_{\mathrm{in}}\int_{\mathrm{NL}}|\mathbf{e}_1|^4 d^3r. \tag{S29}$$

By reciprocity, the field that a nonlinear polarisation $\delta\mathbf{P}$ radiates into the transmitted wave is proportional to $\int \mathbf{e}_1^{\mathrm{rec}}\cdot\delta\mathbf{P}\, d^3r$, where $\mathbf{e}_1^{\mathrm{rec}}$ is the linear field excited by a wave incident from the detection side [26]. The projections above set the measured cascaded and native phases when $\mathbf{e}_1^{\mathrm{rec}}$ is proportional to $\mathbf{e}_1^*$ in the second-order material, as for a film much thinner than the wavelength and symmetric under reflection through its mid-plane, in which $\mathbf{e}_1$ has a uniform phase. A substrate breaks this symmetry; then one factor $\widetilde{M}^*$ of the cascaded projection and one factor $\mathbf{e}_1^*$ of the native projection are replaced by the corresponding integrals with $\mathbf{e}_1^{\mathrm{rec}}$. If the transmitted field has a phase relative to $\mathbf{e}_1$, the real and imaginary parts of the cascaded response mix in that phase. The ratio of the real parts is

$$R^{(1)} = \frac{2\omega_1|\chi^{(2)}|^2\eta^{(1)}}{3n_2^2\chi^{(3)}_{\mathrm{nat}}}\,\frac{|\Delta\omega|}{\Delta\omega^2 + (\gamma_2/2)^2}, \qquad \eta^{(1)} = \frac{\varepsilon_0 n_2^2|\widetilde{M}|^2}{|\chi^{(2)}|^2N_2\int_{\mathrm{NL}}|\mathbf{e}_1|^4 d^3r} \le 1, \tag{S30}$$

which reduces at $\Delta\omega = 2\gamma_2$ to Eq. (6) of the main text with η_{geom} replaced by $\eta^{(1)}$. The local field enhancement enters $\widetilde{M}$ and $\int_{\mathrm{NL}}|\mathbf{e}_1|^4$ alike and cancels in $R^{(1)}$; it enters the nonlinear phase per unit incident power.

For a given design, $\eta^{(1)}$ follows from two full-wave calculations of the unit cell with periodic boundary conditions. An eigenfrequency calculation near $2\omega_1$ gives the second-harmonic mode $\mathbf{f}_2$, its quality factor $Q_2 = \omega_2/\gamma_2$ and N_2 from Eq. Eq. (S2). A frequency-domain calculation at ω_1 with a normally incident plane wave of unit amplitude gives $\mathbf{e}_1$. The overlap $\widetilde{M}$ and $\int_{\mathrm{NL}}|\mathbf{e}_1|^4 d^3r$ are then integrated over the second-order material, with the tensor of Eq. Eq. (S1) rotated to the crystal orientation of the film. Since $\eta^{(1)}$ does not depend on the normalisation of $\mathbf{f}_2$ or on the incident amplitude, neither has to be fixed in the simulations.

The parameters of the design and the Z-scan model.

8. MICRORING

A. Photon-number form

The $\chi^{(2)}$ microresonator literature uses photon-number amplitudes. With $\hat{a} = a_1(2N_1/\hbar\omega_1)^{1/2}$ and $\hat{b} = a_2(N_2/\hbar\omega_1)^{1/2}$, so that $|\hat{a}|^2$ and $|\hat{b}|^2$ are the photon numbers of the pump and second-harmonic modes, Eqs. (1) and (2) of the main text become

$$\dot{\hat{a}} = (i\delta_1 - \kappa_1)\hat{a} - 2ig\,\hat{a}^*\hat{b} + iU_{\rm nat}|\hat{a}|^2\hat{a} + \sqrt{2\kappa_1^{\rm e}}\,s_{\rm in}, \tag{S31}$$

$$\dot{\hat{b}} = (i\Delta\omega - \kappa_2)\hat{b} - ig\,\hat{a}^2, \tag{S32}$$

where $\kappa_n = \gamma_n/2$ are amplitude decay rates, $\kappa_1^{\rm e} = \gamma_1^{\rm e}/2$ refers to the bus waveguide, $|s_{\rm in}|^2$ is the incident photon flux, and the phase of $\hat{b}$ is chosen to make g real. The coupling rate is

$$g = \frac{\varepsilon_0\omega_1|M|}{2N_1N_2^{1/2}}\,(\hbar\omega_1)^{1/2}, \tag{S33}$$

which is the standard rate

$$\Big(\frac{\hbar\omega_1^2\omega_2}{8\varepsilon_0}\Big)^{1/2}\frac{\big|\int\chi^{(2)}_{ijk}E_{1,i}E_{1,j}E^*_{2,k}\,d^3r\big|}{\int\varepsilon_r|\mathbf{E}_1|^2d^3r\,\big(\int\varepsilon_r|\mathbf{E}_2|^2d^3r\big)^{1/2}}$$

of the Hamiltonian $\hat{H}/\hbar = g[(\hat{a}^\dagger)^2\hat{b} + \hat{a}^2\hat{b}^\dagger]$ for a nondispersive resonator [27, 28]. The native Kerr shift per photon is $U_{\rm nat} = \alpha_3\hbar\omega_1/2N_1$. For a ring of radius $R_{\rm ring}$ and mode area $A_{\rm eff}$, the round-trip Kerr phase gives

$$U_{\rm nat} = \frac{\hbar\omega_1^2cn_2^I}{n_{g1}^2V_{\rm eff}}, \qquad V_{\rm eff} = 2\pi R_{\rm ring}A_{\rm eff}, \tag{S34}$$

with n_{g1} the group index at the pump.

B. Cascaded Kerr coefficient and the coupling rate

The steady state of Eq. Eq. (S32), $\hat{b} = -ig\hat{a}^2/(\kappa_2 - i\Delta\omega)$, turns the coupling term of Eq. Eq. (S31) into $(iU_{\rm casc} - \kappa_{\rm casc})|\hat{a}|^2\hat{a}$ with

$$U_{\rm casc} = -\frac{2g^2\Delta\omega}{\Delta\omega^2 + \kappa_2^2}, \qquad \kappa_{\rm casc} = \frac{2g^2\kappa_2}{\Delta\omega^2 + \kappa_2^2}, \tag{S35}$$

the photon-number form of Eq. Eq. (S8). The pump resonance moves to $\omega_a - (U_{\rm nat} + U_{\rm casc})|\hat{a}|^2$, and $R = |U_{\rm casc}|/U_{\rm nat}$. Since g^2 and $U_{\rm nat}$ both scale as $1/V_{\rm eff}$, R does not depend on the size of the ring, and $R \propto Q_2$ at fixed $\Delta\omega/\gamma_2$. These expressions describe the cascaded Kerr response measured in AlN microrings [29, 30].

The coupling rate follows from the second-harmonic efficiency. On double resonance and for an undepleted pump, Eqs. (S31) and (S32) give

$$\eta_{\rm SHG} = \frac{P_{\rm SH}}{P_{\rm in}^2} = \frac{16g^2\kappa_2^{\rm e}(\kappa_1^{\rm e})^2}{\hbar\omega_1\kappa_2^2\kappa_1^4}, \tag{S36}$$

with $P_{\rm in}$ the pump power in the bus waveguide, $P_{\rm SH}$ the second-harmonic power coupled out of the ring and $\kappa_2^{\rm e}$ the out-coupling rate of the second harmonic. This is Eq. (4) of Ref. [28] at zero detuning, and Eq. Eq. (S33) follows Ref. [31].

C. Cascaded shift from measured quantities

The pump loses energy to the second-harmonic mode at the rate $2\kappa_{\rm casc}|\hat{a}|^2$, so $\kappa_{\rm casc}|\hat{a}|^2 = P_{\rm SH}^{\rm tot}/2W_1$, where $W_1 = \hbar\omega_1|\hat{a}|^2$ is the energy stored in the pump mode and $P_{\rm SH}^{\rm tot} = (\kappa_2/\kappa_2^{\rm e})P_{\rm SH}$ is the total second-harmonic power generated, of which $P_{\rm SH}$ leaves through the bus. With $-U_{\rm casc}/\kappa_{\rm casc} = \Delta\omega/\kappa_2$ from Eq. Eq. (S35), the cascaded shift of the pump resonance is

$$\delta\omega_{\rm casc} = -U_{\rm casc}|\hat{a}|^2 = \frac{\Delta\omega}{\kappa_2}\,\frac{P_{\rm SH}^{\rm tot}}{2W_1}. \tag{S37}$$

This relation holds at any pump detuning and power within the elimination, and the same argument applied to Eq. Eq. (S8) gives it for any resonator. For a pump locked to its resonance and with the nonlinear loss neglected, $W_1 = 2\kappa_1^e P_{in}/\kappa_1^2$ and $P_{SH} = \eta_{SHG} P_{in}^2 \kappa_2^2/(\Delta\omega^2 + \kappa_2^2)$ from Eq. Eq. (S36), so

$$\frac{\delta\omega_{casc}}{\kappa_1} = \frac{\eta_{SHG} P_{in}}{4} \frac{\kappa_1}{\kappa_1^e} \frac{\kappa_2}{\kappa_2^e} \frac{\kappa_2 \Delta\omega}{\Delta\omega^2 + \kappa_2^2}, \tag{S38}$$

which is Eq. (7) of the main text at critical coupling, $\kappa_n^e = \kappa_n/2$. The coupling ratios follow from the loaded and intrinsic quality factors, $\kappa_n^e/\kappa_n = 1 - Q_n^{load}/Q_n^{int}$. For the device of Ref. [28] they are 0.70 and 0.22, the product of the coupling factors in Eq. Eq. (S38) is 1.59, and $\eta_{SHG} = 5 \times 10^6$%/W gives $\delta\omega_{casc}/\kappa_1 = 0.094$ at $P_{in} = 5$ µW and $\Delta\omega = 2\gamma_2$. With the pump locked, the nonlinear loss lowers the photon number by the factor $(1 + \kappa_{casc}|\hat{a}|^2/\kappa_1)^{-2}$, which is 0.96 at this point and 0.76 at $|\Delta\omega| = \kappa_2$, where $\rho = 1$ [main text, Fig. 5(b)]. Measuring P_{SH} and W_1 at the operating point and using Eq. Eq. (S37) would avoid this correction.

D. Design points

Of the three designs in Table S2, design A is the periodically poled ring of Ref. [28], with the coupling rate obtained there from a fit of the measured lineshape. Design B applies the same poling to a ring with a higher pump quality factor; its g is scaled from A as $V_{eff}^{-1/2}$. Design C is ring B without poling, phase matched through mode order or the cyclic crystal orientation, for which we take g 12 times smaller, an estimate for an interaction through d_{31} with a mismatched transverse mode. For design A, Eq. Eq. (S36) with the efficiency reported in Ref. [28] gives $g/2\pi = 0.56$ MHz instead of the fitted 1.2 MHz, which would lower its R to about 600; the fit may include a residual second-harmonic detuning that Eq. Eq. (S36) omits. Since U_{nat} uses the assumed A_{eff} while g is measured, R of design A also scales with A_{eff}.

Table S2. Microring Designs and Cascaded Response at $\Delta\omega = 2\gamma_2$

Design	Q_1	Q_2	R_{ring} (µm)	$g/2\pi$ (MHz)	R	$U/2\pi$ (Hz)	P_{bist} (µW)	P_{par} (µW)
A, poled [28]	5.4×10^5	4.5×10^5	70	1.20	2800	−1590	73	19
B, poled	1.0×10^6	3.0×10^5	50	1.42	1830	−1470	37	8.0
C, not poled	1.0×10^6	3.0×10^5	50	0.12	13	−10	7000	1150
B, native only	1.0×10^6	–	50	–	–	+0.80	14400	9400

Loaded quality factors; the intrinsic values are 1.8×10^6 and 5.8×10^5 for A and 2×10^6 and 6×10^5 for B and C. Pump wavelength 1.56 µm (A) and 1.55 µm (B, C); $A_{eff} = 1$ µm^2, $n_{g1} = 2.30$, $n_{g2} = 2.45$, $n_2^I = 1.8 \times 10^{-19}$ m^2/W. U is U_{casc}, or U_{nat} in the native row. P_{bist} and P_{par} are input powers at the bistability and parametric thresholds of Sections 8.E and 8.F, for a cold mismatch of $2\gamma_2$ and a second-order dispersion $D_2/2\pi = 10$ MHz.

E. Bistability

The steady state of Eq. Eq. (S31) with the cascaded terms is the cubic of Section 4.C in photon units, with $\alpha' \to U_{nat} + U_{casc}$, $\alpha'' \to \kappa_{casc}$, $\gamma_1/2 \to \kappa_1$ and the drive $2\kappa_1^e|s_{in}|^2$. In a ring the pump detuning needed for bistability, a few κ_1, changes $\Delta\omega$ by an amount comparable with κ_2, because $\Delta\omega = \Delta\omega_0 + 2\delta_1$ with $\Delta\omega_0 = 2\omega_a - \omega_2$ fixed by the device. The threshold is therefore found by minimising the fold power over δ_1 with U_{casc} and κ_{casc} evaluated at $\Delta\omega_0 + 2\delta_1$. The condition $\rho < 1/\sqrt{3}$ of Section 4.C then applies to the pump-frame $\Delta\omega$ and does not restrict $\Delta\omega_0$. For design B the threshold is 37 µW at $\Delta\omega_0 = 2\gamma_2$ and has a shallow minimum of 32 µW at $\Delta\omega_0 = 3.3\gamma_2$, against 14.4 mW for the native response of this ring. These are the folds of the single-mode cubic. In a multimode ring the parametric threshold of Section 8.F lies below them, for the cascaded and the native response alike, so the upper branch carries parametric sidebands unless the dispersion suppresses them. A shift of the pump resonance by its half-width κ_1 needs $\kappa_1/|U_{nat} + U_{casc}| = 6.6 \times 10^4$ photons in design B, an intracavity energy of 8.4 fJ, against 15 pJ for the native response.

The steady state of the two-mode equations stays stable up to a cascaded shift of about $1.1|\Delta\omega|$. For design B at $\Delta\omega = 2\gamma_2$ and $\delta_1 = 0$, the largest real part of the eigenvalues of the four-dimensional Jacobian of Eqs. (S31) and (S32) crosses zero at 2.0×10^6 photons, a cascaded shift of $30\kappa_1$. This photon number is 1.1 times $(\Delta\omega^2 + \kappa_2^2)/2g^2$, the value at which $|U_{casc}||\hat{a}|^2 = |\Delta\omega|$.

F. Parametric gain

The degenerate four-wave mixing that produces parametric gain is mediated by the same second-harmonic mode b_0 as the pump self-phase modulation, while the cross-phase terms of the sidebands pass through the second-harmonic modes $b_\pm$ at azimuthal orders $2m_p \pm q$. With the pump a_0 at order m_p and the sidebands $a_\pm$ at $m_p \pm q$, the relevant part of the Hamiltonian is

$$\hat{H}/\hbar \supset g\,\hat{a}_0^{\dagger 2}\hat{b}_0 + 2g\,\hat{a}_+^\dagger \hat{a}_-^\dagger \hat{b}_0 + 2g\,\hat{a}_0^\dagger \hat{a}_+^\dagger \hat{b}_+ + 2g\,\hat{a}_0^\dagger \hat{a}_-^\dagger \hat{b}_- + \text{h.c.}, \tag{S39}$$

assuming equal overlaps for all triplets. Linearising about the pump gives a four-dimensional system for $(\delta\hat{a}_+, \delta\hat{a}_-^*, \delta\hat{b}_+, \delta\hat{b}_-^*)$, and the threshold is the lowest input power at which an eigenvalue acquires a positive real part, minimised over δ_1 and q. The modes $b_\pm$ are detuned from b_0 by $\pm q(D_1^{(2)} - D_1^{(1)})$, where $D_1^{(n)}$ is the free spectral range of band n in angular units and $D_1^{(2)} - D_1^{(1)} = D_1^{(1)}(n_{g1}/n_{g2} - 1)$, $-2\pi \times 25$ GHz or $-20\gamma_2$ per azimuthal order for design B. The sidebands are detuned from the pump-band modes by the integrated dispersion $D_2 q^2/2$, with D_2 the second-order dispersion parameter. Without the walk-off of the second-harmonic modes the system reduces to the single-pair result of Section 4.D. For design B the threshold is 8.0 µW, with a spread below 2% for anomalous dispersion $D_2/2\pi$ between 1 and 50 MHz, and 11.6 µW for normal dispersion, $D_2/2\pi = -10$ MHz, against 9.4 mW for the native response.

G. Response time

In a ring $\gamma_2/\gamma_1 = 2Q_1/Q_2$, which is 6.7 for design B, so the second-harmonic mode responds faster than the pump mode. We compare the small-signal response of Eqs. (S31) and (S32) to a modulation of the input power with that of a single-mode cavity with the same photon number, Kerr coefficient and nonlinear loss but an instantaneous response. For design B at $\Delta\omega = 2\gamma_2$, $\delta_1 = 0$ and 7 µW, below the parametric threshold, the 3-dB bandwidth is smaller by 4%, and by 7% at 20 µW.

H. Thermal and photorefractive shifts

At the powers of Table S2, absorption heats the ring and shifts the pump resonance to lower frequency. In a lumped model the shift is $|d\omega_a/dT| R_{\text{th}} P_{\text{abs}}$, with R_{th} the thermal resistance of the ring and

$$P_{\text{abs}} = f_{\text{abs}}\left[2\hbar\omega_1\kappa_{1,0}|\hat{a}|^2 + \frac{\kappa_{2,0}}{\kappa_2}\,2\hbar\omega_1\kappa_{\text{casc}}|\hat{a}|^4\right], \tag{S40}$$

where $\kappa_{n,0}$ are intrinsic decay rates and f_{abs} is the absorbed fraction of the intrinsic loss. The second term is the power transferred to the second harmonic, of which the fraction $\kappa_{2,0}/\kappa_2$ stays on chip. For design B at 7 µW with the pump at the cold resonance, $dn/dT = 3.3 \times 10^{-5}$ K^{-1}, $R_{\text{th}} = 3 \times 10^4$ K/W and $f_{\text{abs}} = 0.3$, the thermal shift is $1.1\kappa_1$ against a cascaded shift of $0.7\kappa_1$, in line with the thermal shifts measured in thin-film $LiNbO_3$ rings [32]. The photorefractive effect lowers the second-harmonic efficiency of poled thin-film rings above about 10 µW [28] and acts on $\Delta\omega$; it is reduced by removing the cladding [33].

The thermal shift depends on $\Delta\omega$ through κ_{casc}, which is even, and the photorefractive charge generation follows the second-harmonic intensity, also even to leading order, whereas U_{casc} is odd. To the extent that the backgrounds are even, the cascaded shift is

$$\delta\omega^{\text{casc}}(\Delta\omega) = \tfrac{1}{2}\left[\delta\omega(\Delta\omega) - \delta\omega(-\Delta\omega)\right] \tag{S41}$$

under two conditions. First, the pump must be locked to the resonance as shifted by all mechanisms. The photon number then depends on $\Delta\omega$ only through κ_{casc}, which is even, and so do the thermal and photorefractive shifts. A pump held at a fixed frequency does not meet this condition, because the odd cascaded shift and the even thermal shift detune the cavity differently for $\pm\Delta\omega$; with the thermal shift of the previous paragraph, Eq. Eq. (S41) then returns less than half of the cascaded shift. Second, the zero of the pump-frame $\Delta\omega$ must be referenced in the same run to the maximum of the second-harmonic power, since an even shift of the pump frequency or a photorefractive offset of $\Delta\omega$ mixes the odd and even parts. Repeating the measurement at two powers would test for a residual odd background, since the cascaded shift is linear in the photon number while photorefractive saturation is not.

I. Electro-optic and thermal tuning of $\Delta\omega$

The electro-optic effect shifts each resonance by $\omega_n n_n^3 r_n E_{\mathrm{dc}}/(2n_{gn})$ for a static field E_{dc} along the optic axis, so

$$\frac{d\Delta\omega_0}{dE_{\mathrm{dc}}} = \omega_1\Big(\frac{n_1^3 r_1}{n_{g1}} - \frac{n_2^3 r_2}{n_{g2}}\Big) \tag{S42}$$

for the cold mismatch $\Delta\omega_0$, with r_n the electro-optic coefficient $r_{33} \approx 31$ pm/V [34] at each wavelength. The two terms nearly cancel, so the rate is set by the dispersion of the indices and of r_{33}. With the extraordinary indices of Table S1, the group indices of Table S2 and equal r_{33} at both wavelengths it is $+2\pi \times 0.2$ GHz per V/µm; an r_{33} larger by 10% at 775 nm gives $-2\pi \times 2.4$ GHz per V/µm, or $1.8\gamma_2$ for design B. The rate therefore has to be calibrated on the device. The pump resonance itself moves by $2\pi \times 13$ GHz per V/µm, about $130\kappa_1$, so a sweep of $\Delta\omega$ over $\pm 2\gamma_2$, which needs 2.2 V/µm peak to peak, requires the pump to follow the resonance. The difference of the thermo-optic shifts of the two modes also tunes $\Delta\omega$, by γ_2 for a change of about 1.4 K in our estimate, with a magnitude and sign that depend on the dispersion of dn/dT and must be measured. An electro-optic dither of $\Delta\omega$ with lock-in detection and the pump locked to the resonance would extract the odd part of Eq. Eq. (S41) faster than the thermal and photorefractive responses.

9. PARAMETERS

Material constants, mode parameters and design points used in all figures and tables.

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